Assume/Guarantee Contract of `Fg333saUnPack`

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1 Variable Declaration

Maps : `variable` in code $\mapsto$ variable in formal specification

Inputs :

- `*pRecvbuf` $\mapsto$ pRecvbuf : $\mathbb{N}^*$

where $\mathbb{N}^* = \bigcup_{n\in\mathbb{N}} \mathbb{N}^n$.

Outputs :

- `gf` $\mapsto$ gf : $\mathbb{R}$
- `wt0` $\mapsto$ wt0 : $\mathbb{R}$
- `glwd1` $\mapsto$ glwd1 : $\mathbb{N}$
- `glwd2` $\mapsto$ glwd2 : $\mathbb{N}$
- `dlwd` $\mapsto$ dlwd : $\mathbb{N}$
- `status` $\mapsto$ status : $\mathbb{N}$

In-Outputs : None

States : None

Parameters :

- `dtime` $\mapsto$ dtime : $\mathbb{R}$

2 AG Contract

Assumptions :

$$Assumptions \triangleq |\mathsf{pRecvbuf}| \geq 17$$

Guarantees :

$$Guarantees \triangleq \exists \mathsf{pbuff}' \cdot (Assign_1 \wedge Assign_2 \wedge Assign_3 \wedge Assign_4 \wedge Assign_5 \wedge Assign_6)$$

$$Assign_1 \triangleq \mathsf{wt0} := \frac{2^{12}\mathsf{pRecvbuf}_2 + 2^8\mathsf{pRecvbuf}_3 + 2^4\mathsf{pRecvbuf}_4 + \mathsf{pRecvbuf}_5}{3 \cdot 10^6 \cdot 0.0174532925199433}$$

$$Assign_2 \triangleq \mathsf{glwd1} := 2^4 \cdot \mathsf{pRecvbuf}_6 + \mathsf{pRecvbuf}_7$$

$$Assign_3 \triangleq \mathsf{glwd2} := 2^4 \cdot \mathsf{pRecvbuf}_8 + \mathsf{pRecvbuf}_9$$

$$Assign_4 \triangleq \mathsf{dlwd} := 2^4 \cdot \mathsf{pRecvbuf}_{10} + \mathsf{pRecvbuf}_{11}$$

$$Assign_5 \triangleq \mathsf{status} := \mathsf{pRecvbuf}_{16}$$

$$Assign_6 \triangleq \mathsf{gf} := \mathsf{wt0}' \cdot \mathsf{dtime}$$